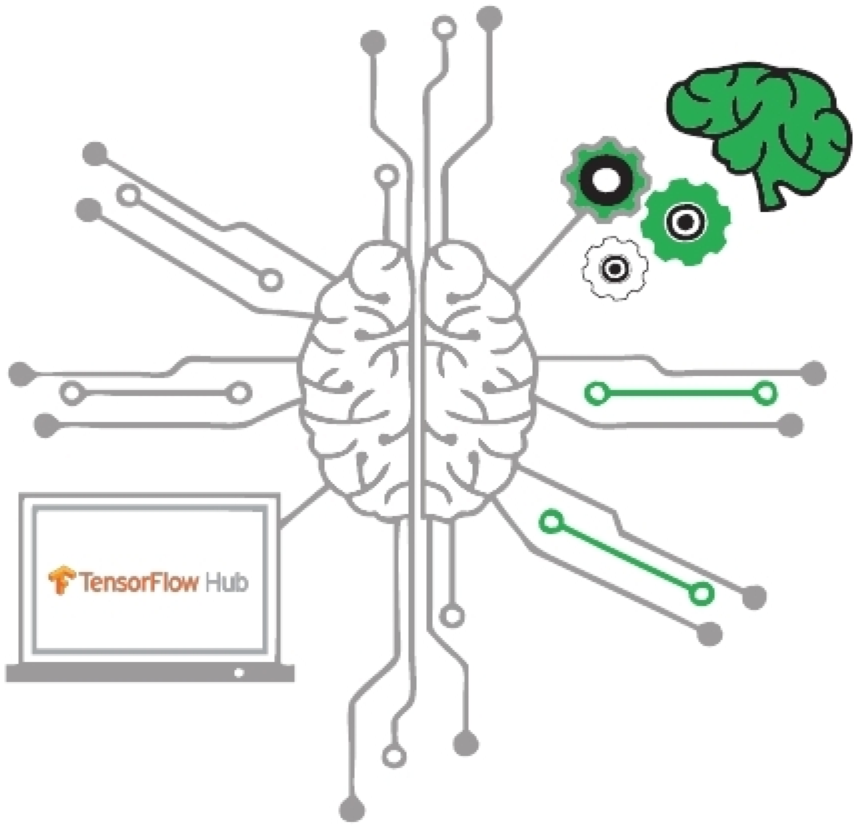
TensorFlow Hub: A Machine Learning Ecosystem



Machine learning (ML) is the new 'in' thing in the world of computer science. For a newbie to get into it is not very easy, however. TensorFlow Hub is one way in which newbies can work with tried and tested models, and hone their skills in this rapidly evolving field.

Starting from scratch, machine learning requires four key components — algorithms, data, compute, and expertise. If you don't have access to any one of these, then good luck to getting a great result! One viable alternative for small-scale projects is to reproduce pre-existing experiments relevant to their use cases, tweak them, and improve on the results. However, that is easier said than done.

Reproducibility in machine learning is a crisis that has been highlighted by top researchers and publications over the years. Recent times have seen a shift towards a more open paradigm. The field's largest conference, NeurIPS, received 8,000 submissions last year, with organisers now encouraging code-sharing for papers submitted to them. This has been done in full consideration of existing proprietary libraries and dependencies that cannot accompany the code-release, i.e., code submission is encouraged even if it is 'non-executable' in external environments. The move has been widely perceived as a start to a more open culture in solving problems, working with data sets, and in general, promoting transparency and reproducibility.

"Machine learning is really hard to do, so we built TensorFlow Hub (TFHub) to allow people to reuse machine learning. We are letting people pack up good machine learning and

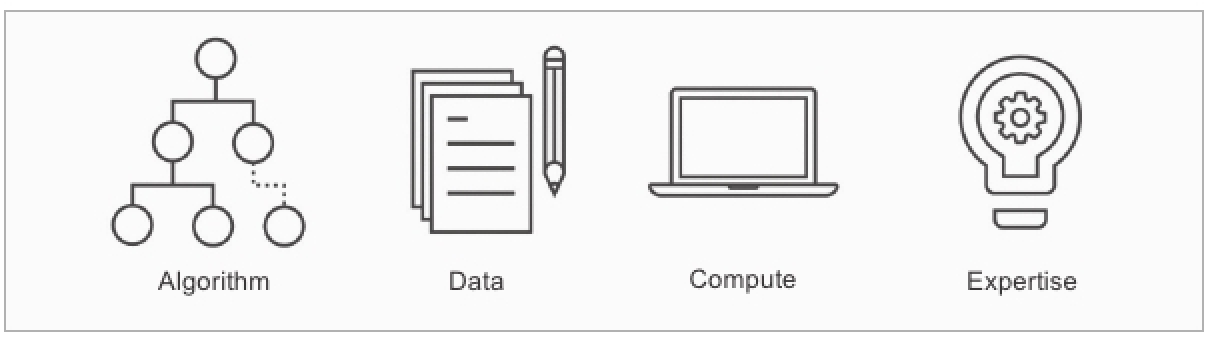


Figure 1: Prerequisites for machine learning [Source: Google Inc.]

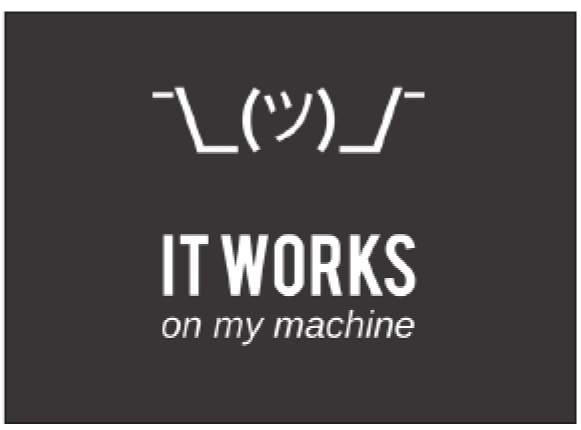


Figure 2: The most overused developer excuse for reproducibility! [Source: teepublic.com]

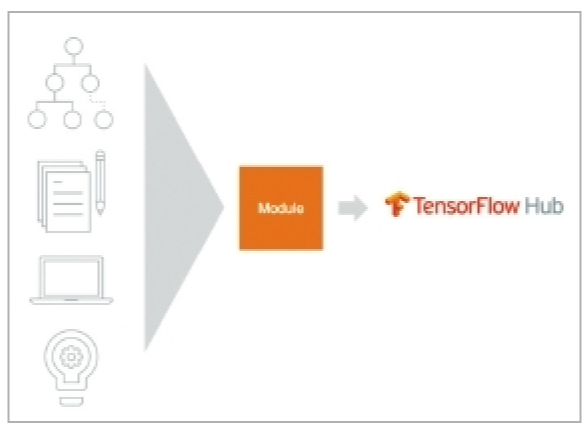


Figure 3: An overview of TFHub's functionality [Source: Google Inc.]